

12-group: 群论 (Group Theory)

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2026 年 06 月 09 日



*“In mathematics, more specifically algebra, **abstract algebra** or **modern algebra** is the study of **algebraic structures**, which are **sets** with specific **operations** acting on their elements.”*

— Wikipedia

*“Algebraic structures include **groups**, rings, fields, modules, vector spaces, lattices, and algebras over a field.”*

15-251: Great Theoretical Ideas in Computer Science
Spring 2016, Lecture 21

Group Theory



“论五次方程的代数解法问题” (1829)



Augustin-Louis Cauchy
(1789 ~ 1857)



Joseph Fourier
(1768 ~ 1830)



Siméon Denis Poisson
(1781 ~ 1840)

“Ask **Jacobi** or **Gauss** publicly to give their opinion,
not as to the **truth**, but as to the **importance** of these theorems.”

“Is there a formula for the roots of a ≥ 5 degree polynomial equation in terms of its coefficients, using only $+$, $-$, $\times$, $\div$, $\sqrt[r]{}$?”

$$ax^2 + bx + c = 0 (a \neq 0)$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x^3 + px + q = 0$$

直接给出公式。对于一元三次方程 $ax^3 + bx^2 + cx + d = 0$ ，卡尔丹公式求根的表达式为

$$x_1 = \left[\sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} \right] - \frac{b}{3a}$$

$$x_2 = \left[\omega \cdot \sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + \omega^2 \cdot \sqrt[3]{-\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} \right] - \frac{b}{3a}$$

$$x_3 = \left[\omega^2 \cdot \sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + \omega \cdot \sqrt[3]{-\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} \right] - \frac{b}{3a}$$

其中:

$$p = \frac{3ac - b^2}{3a^2}, \quad q = \frac{2b^3 - 9abc + 27a^2d}{27a^3}, \quad \omega = e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$



Girolamo Cardano
(1501 ~ 21/09/1576)

对于一元四次方程

$$ax^4 + bx^3 + cx^2 + dx + e = 0$$

记

$$\begin{cases} \Delta_1 = c^2 - 3bd + 12ae \\ \Delta_2 = 2c^3 - 9bcd + 27ad^2 + 27b^2e - 72ace \end{cases}$$

并记

$$\Delta = \frac{\sqrt[3]{2}\Delta_1}{3a\sqrt[3]{\Delta_2 + \sqrt{-4\Delta_1^3 + \Delta_2^2}}} + \frac{\sqrt[3]{\Delta_2 + \sqrt{-4\Delta_1^3 + \Delta_2^2}}}{3\sqrt[3]{2}a}$$

则有

$$\begin{cases} x_1 = -\frac{b}{4a} - \frac{1}{2}\sqrt{\frac{b^2}{4a^2} - \frac{2c}{3a}} + \Delta - \frac{1}{2}\sqrt{\frac{b^2}{2a^2} - \frac{4c}{3a} - \Delta - \frac{-\frac{b^3}{a^3} + \frac{4bc}{a^2} - \frac{8d}{a}}{4\sqrt{\frac{b^2}{4a^2} - \frac{2c}{3a}} + \Delta}} \\ x_2 = -\frac{b}{4a} - \frac{1}{2}\sqrt{\frac{b^2}{4a^2} - \frac{2c}{3a}} + \Delta + \frac{1}{2}\sqrt{\frac{b^2}{2a^2} - \frac{4c}{3a} - \Delta - \frac{-\frac{b^3}{a^3} + \frac{4bc}{a^2} - \frac{8d}{a}}{4\sqrt{\frac{b^2}{4a^2} - \frac{2c}{3a}} + \Delta}} \\ x_3 = -\frac{b}{4a} + \frac{1}{2}\sqrt{\frac{b^2}{4a^2} - \frac{2c}{3a}} + \Delta - \frac{1}{2}\sqrt{\frac{b^2}{2a^2} - \frac{4c}{3a} - \Delta + \frac{-\frac{b^3}{a^3} + \frac{4bc}{a^2} - \frac{8d}{a}}{4\sqrt{\frac{b^2}{4a^2} - \frac{2c}{3a}} + \Delta}} \\ x_4 = -\frac{b}{4a} + \frac{1}{2}\sqrt{\frac{b^2}{4a^2} - \frac{2c}{3a}} + \Delta + \frac{1}{2}\sqrt{\frac{b^2}{2a^2} - \frac{4c}{3a} - \Delta + \frac{-\frac{b^3}{a^3} + \frac{4bc}{a^2} - \frac{8d}{a}}{4\sqrt{\frac{b^2}{4a^2} - \frac{2c}{3a}} + \Delta}} \end{cases}$$

Theorem (Abel-Ruffini Theorem)

*There is **no** solution in **radicals** to polynomial equations of ≥ 5 degree.*



Niels Henrik Abel (1802 ~ 1829)

Theorem (Galois Theorem)

*An equation is **solvable** in terms of radicals **if and only if** the **Galois group** of its splitting field is **solvable**.*

近世代数

群论 (一)

孙智伟 南京大学 教授



https://www.bilibili.com/video/BV1Ex411k7wk?share_source=copy_web

“我看出了 *Galois* 用来证明这个美妙定理的方法是完全正确的。
在那个瞬间,我体验到一种强烈的愉悦。”

— *J. Liouville* (刘维尔; 1846)

Definition (Group (群))

A **group** $(G, *)$ is a **set** G together with a **binary operation** $*$ such that the following four **group axioms** are satisfied:

Closure (封闭):

$$\forall a, b \in G \quad a * b \in G$$

Associativity (结合律):

$$\forall a, b, c \in G \quad (a * b) * c = a * (b * c)$$

Identity (单位元):

$$\exists e \in G \quad \forall a \in G \quad e * a = a \wedge a * e = a$$

Inverse (逆元): Let e be **the** identity of G .

$$\forall a \in G \quad \exists b \in G \quad a * b = e \wedge b * a = e$$

The inverse of a is denoted a^{-1} .

$$\forall n \in \mathbb{Z}^+. a^n \triangleq \underbrace{a * a * \cdots * a}_{\# = n}$$

$$a^0 \triangleq e$$

$$a^{-n} \triangleq (a^{-1})^n$$

Definition (Commutative Group (交换群); Abelian Group (阿贝尔群))

Let $(G, *)$ be a group. If $*$ is commutative,

$$\forall a, b \in G. a * b = b * a,$$

then $(G, *)$ is a commutative group.

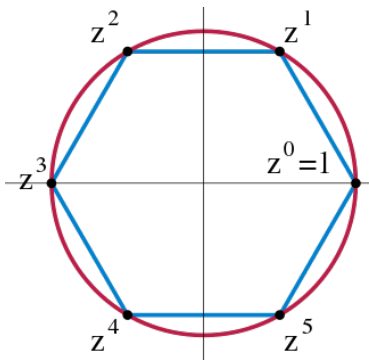
$$(\mathbb{Z}, +)$$

$$(\mathbb{Q} \setminus \{0\}, \times)$$

$$(\{1, -1, \mathbf{i}, -\mathbf{i}\}, \times)$$

Group of n -th Roots of Unity (n 次单位根群)

$$\begin{aligned} U_n &= \{z \in \mathbb{C} \mid z^n = 1\} \\ &= \left\{ \cos \frac{2k\pi}{n} + \mathbf{i} \sin \frac{2k\pi}{n} \mid k = 0, 1, \dots, n-1 \right\} \end{aligned}$$



Quaternion Group (四元数群)

$$(1, i, j, k, -1, -i, -j, -k)$$

x	e	$\bar{e}$	i	$\bar{i}$	j	$\bar{j}$	k	$\bar{k}$
e	e	$\bar{e}$	i	$\bar{i}$	j	$\bar{j}$	k	$\bar{k}$
$\bar{e}$	$\bar{e}$	e	$\bar{i}$	i	$\bar{j}$	j	$\bar{k}$	k
i	i	$\bar{i}$	$\bar{e}$	e	k	$\bar{k}$	$\bar{j}$	j
$\bar{i}$	$\bar{i}$	i	e	$\bar{e}$	$\bar{k}$	k	j	$\bar{j}$
j	j	$\bar{j}$	$\bar{k}$	k	$\bar{e}$	e	i	$\bar{i}$
$\bar{j}$	$\bar{j}$	j	k	$\bar{k}$	e	$\bar{e}$	$\bar{i}$	i
k	k	$\bar{k}$	j	$\bar{j}$	$\bar{i}$	i	$\bar{e}$	e
$\bar{k}$	$\bar{k}$	k	$\bar{j}$	j	i	$\bar{i}$	e	$\bar{e}$



Cayley Table

$$i^2 = j^2 = k^2 = 1 \quad ij = k, ji = -k, jk = i, kj = -i, ki = j, ik = -j$$

Theorem

Let G be a group.

- (1) *The identity is unique.*
- (2) *The inverse of each element is unique.*
- (3) $\forall a \in G. (a^{-1})^{-1} = a.$
- (4) $\forall a, b \in G. (ab)^{-1} = b^{-1}a^{-1}.$
- (5) $\forall a, b, c \in G. (ab = ac \implies b = c) \wedge (ba = ca \implies b = c).$
- (6) $\forall a, b \in G. \exists! x \in G. ax = b \wedge ya = b.$

Additive Group of Integers Modulo m (模 m 剩余类加群)

$$(\mathbb{Z}_m = \{0, 1, \dots, m-1\}, +_m)$$

$$(\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}, \times_6)$$

Multiplicative Group of Integers Modulo m (模 m 剩余类乘法群)

$$U(m) = \{a \in \mathbb{Z}_m \mid (a, m) = 1\}$$

Theorem (Bézout's Identity)

$$(a, b) = d \implies \exists u, v \in \mathbb{Z}. au + bv = d$$

$$(a, m) = 1 \implies \exists u, v \in \mathbb{Z}. au + mv = 1$$

$$a^{-1} = u$$

$$(u, m) = 1 \quad ua = au = au + mv = 1 \pmod{m}$$

When p is a prime,

$$\mathbb{Z}_p^* \triangleq U(p) = \{1, 2, \dots, p-1\}$$

Multiplicative Group of Integers Modulo m (模 m 剩余类乘法群)

$$U(m) = \{a \in \mathbb{Z}_m \mid (a, m) = 1\}$$

$$|U(m)| = \varphi(m)$$

Definition (Euler's Totient Function (1763))

$$\varphi(m) = n \prod_{p|n \wedge p \text{ is a prime}} \left(1 - \frac{1}{p}\right)$$

$$\varphi(20) = 20\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{5}\right) = 8$$

$$U(20) = \{1, 3, 7, 9, 11, 13, 17, 19\}$$

Theorem

Let G be an abelian group of *order* n .

$$\forall a \in G. a^n = e.$$

$$G = \{a_1, a_2, \dots, a_n\}$$

$$aG \triangleq \{aa_1, aa_2, \dots, aa_n\} = G$$

$$\prod_{i=1}^n (aa_i) = \prod_{i=1}^n a_i$$

$$a^n a_1 \dots a_n = a_1 \dots a_n \implies a^n = e$$

Theorem (Euler Theorem (1736))

Let $m \in \mathbb{Z}^+$ and $a \in \mathbb{Z}$. If $(a, m) = 1$, then

$$a^{\varphi(m)} \equiv 1 \pmod{m}$$

7^{222} 的最后一位数字?

$$7^{222} \pmod{10}$$

$$(7, 10) = 1 \quad \varphi(10) = 4$$

$$7^{\varphi(10)} = 7^4 \equiv 1 \pmod{10}$$

$$7^{222} \equiv 7^{4 \times 55 + 2} \equiv 7^2 \equiv 9 \pmod{10}$$

Theorem (Euler Theorem (1736))

Let $m \in \mathbb{Z}^+$ and $a \in \mathbb{Z}$. If $(a, m) = 1$, then

$$a^{\varphi(m)} \equiv 1 \pmod{m}$$

$$U_m = \{a \in \mathbb{Z}_m \mid (a, m) = 1\}$$

$$(a, m) = 1 \implies a \in U_m$$

$$a^{\varphi(m)} \equiv 1 \pmod{m}$$

Theorem (Fermat's Little Theorem (1640))

Let p be a prime. Then for any $a \in \mathbb{Z}^+$,

$$a^{p-1} \equiv 1 \pmod{p}$$

$$\varphi(p) = p - 1$$

Definition (Subgroup (子群))

Let $(G, *)$ be a group and $\emptyset \neq H \subseteq G$.

If $(H, *)$ is a group, then we call H a **subgroup** of G , denoted $H \leq G$.

$H = G, H = \{e\}$ are two **trivial** (平凡) subgroups.

If $H \subset G$, then H is a **proper** subgroup (真子群).

$$(H = \{mz \mid z \in \mathbb{Z}\}, +) \leq (\mathbb{Z}, +)$$

$$H = \{1, 2, 4\} \leq G = \mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}$$

Theorem

Suppose that $H \leq G$.

(1) The identity of H is the same with that of G .

$$e_H = e_G$$

(2) The inversion of a in H is the same with that in G .

$$\forall a \in H \ a_H^{-1} = a_G^{-1}$$

$$e_H e_H = e_H = e_H e_G \implies e_H = e_G$$

$$a a_H^{-1} = e_H = e_G = a a_G^{-1} \implies a_H^{-1} = a_G^{-1}$$

Theorem

Let G be a group and $\emptyset \neq H \subseteq G$. $H \leq G$ iff

$$\forall a, b \in H \quad ab^{-1} \in H.$$

$$e = aa^{-1} \in H$$

$$a^{-1} = ea^{-1} \in H$$

$$ab = a(b^{-1})^{-1} \in H$$

Theorem

Suppose that $H_1 \leq G, H_2 \leq G$.

$$H_1 \cap H_2 \leq G.$$

$$H_1 = 2\mathbb{Z} \leq \mathbb{Z} \quad H_2 = 3\mathbb{Z} \leq \mathbb{Z}$$

$$H_1 \cap H_2 = 6\mathbb{Z} \leq \mathbb{Z}$$

$$H_1 \cup H_2?$$

Theorem

Suppose that $H_1 \leq G, H_2 \leq G$.

$$H_1 \cap H_2 \leq G.$$

$$e \in H_1 \cap H_2 \implies H_1 \cap H_2 \neq \emptyset$$

$$\boxed{\forall a, b \in H_1 \cap H_2 \quad ab^{-1} \in H_1 \cap H_2}$$

$$a, b \in H_1 \cap H_2 \implies a, b \in H_1 \wedge a, b \in H_2$$

$$ab^{-1} \in H_1 \wedge ab^{-1} \in H_2$$

Center (中心)

Let G be a group. Let

$$C(G) \triangleq \{g \in G \mid gx = xg, \forall x \in G\}.$$

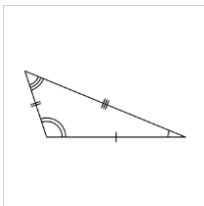
Then $C(G) \leq G$.

$$\forall x \in G \quad ex = xe \implies e \in C(G) \implies C(G) \neq \emptyset$$

$$\boxed{\forall a, b \in C(G) \quad ab^{-1} \in C(G)}$$

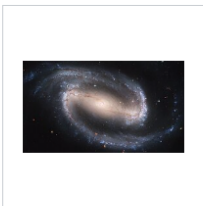
$$\begin{aligned} \forall x \in G \quad (ab^{-1})x &= a(\textcolor{red}{b}^{-1}\textcolor{red}{x}) = a(\textcolor{red}{x}\textcolor{red}{b}^{-1}) \\ &= (ax)b^{-1} = (xa)b^{-1} \\ &= x(ab^{-1}) \end{aligned}$$

Examples of cyclic groups in rotational symmetry



Scalene triangle

C_1



NGC 1300

C_2



Flag of the Isle of Man

C_3



Celtic knot

C_4



Flag of Hong Kong

C_5



Olavsrose [de]

C_6

Definition (Isomorphism (同构))

Let $(G, \cdot)$ and $(G', *)$ be two groups. Let ϕ be a **bijection** such that

$$\forall a, b \in G \quad \phi(a \cdot b) = \phi(a) * \phi(b).$$

Then ϕ is an **isomorphism** from G to G' .

G and G' are isomorphic

$$\phi : G \cong G'$$

$$(\mathbb{R}, +) \cong (\mathbb{R}^+, *)$$

$$\phi(x) = e^x$$

Theorem

Suppose that $\phi : G \cong G'$. Let e and e' are identities of G and G' , respectively.

$$(1) \quad \phi(e) = e'$$

$$(2) \quad \phi(a^{-1}) = (\phi(a))^{-1}$$

$$(3) \quad \phi^{-1} : G' \cong G$$

$$e'\phi(e) = \phi(e) = \phi(ee) = \phi(e)\phi(e) \implies \phi(e) = e'$$

$$\phi(a)\phi(a^{-1}) = \phi(aa^{-1}) = \phi(e) = e'$$

Klein Four-group (四元群; $K_4; V$)

*	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

$$a^2 = b^2 = c^2 = (ab)^2 = e$$

$$ab = c = ba \quad ac = b = ca \quad bc = a = cb$$

$$K_4 \cong U(8) = \{1, 3, 5, 7\}$$

Definition (Order of Elements (元素的阶))

Let G be a group, e be the identity of G .

The **order** of a is the **smallest** positive integer r such that $a^r = e$.

$$\text{ord } a = r$$

If such r does not exist, then $\text{ord } a = \infty$.

$$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$$

$$\mathbb{Z}_5^* = \{1, 2, 3, 4\}$$

$$(\mathbb{Z}, +)$$

Theorem

Let G be a group and e be the identity of G .

$$((\text{ord } a = n) \wedge (\exists m \in \mathbb{Z}. a^m = e)) \implies n \mid m.$$

$$m = nq + r \quad (0 \leq r < n)$$

If $r > 0$,

$$a^r = a^{m-nq} = a^m \cdot (a^n)^{-q} = e \cdot e = e$$

$$\text{ord } a \neq n$$

Theorem

Let G be a group and e be its identity.

$$\text{ord } a = n \implies \forall m \in \mathbb{Z} \text{ ord } a^m = \frac{n}{(n, m)}$$

$$\boxed{\text{ord } a^m = d}$$

$$(a^m)^d = e$$

$$\implies a^{md} = e$$

$$(a^m)^{\frac{n}{(m, n)}} = a^{\frac{mn}{(m, n)}} = (a^n)^{\frac{m}{(m, n)}} = e$$

$$\implies n \mid md$$

$$\implies \frac{n}{(m, n)} \mid \frac{m}{(m, n)} d$$

$$d \mid \frac{n}{(m, n)}$$

$$\implies \frac{n}{(m, n)} \mid d$$

Definition (Cyclic Group (循环群))

Let G be a group. If

$$\exists a \in G \ G = \langle a \rangle \triangleq \{a^r \mid r \in \mathbb{Z}\} = \{\dots, a^{-2}, a^{-1}, a^0 = e, a, a^2, \dots\},$$

then G is a **cyclic group**.

If $G = \langle a \rangle$, then a is a **generator** (生成元) of G .

$(\mathbb{Z}, +)$ is an **infinite** cyclic group

$$(\mathbb{Z}, +) = \langle 1 \rangle = \langle -1 \rangle$$

$(\mathbb{Z}_n, +)$ is a **finite** cyclic group of order n

$$(\mathbb{Z}_m, +) = \langle 1 \rangle$$

$$\mathbb{Z}_5^* = \{1, 2, 3, 4\}$$

$$\mathbb{Z}_5^* = \langle 2 \rangle = \langle 3 \rangle$$

Theorem

(1) Let $G = \{e, a, a^{-1}, a^2, a^{-2}, \dots\}$ be an *infinite* cyclic group.

$$\forall k, l \in \mathbb{Z} (a^k = a^l \rightarrow k = l).$$

(2) Let $G = \{e, a, a^2, \dots, a^{n-1}\}$ be a *finite* cyclic group of order n .

$$\forall k, l \in \mathbb{Z} (a^k = a^l \leftrightarrow n \mid (k - l)).$$

Theorem (Structure Theorem of Cyclic Groups (循环群结构定理))

Let $G = \langle a \rangle$ be a cyclic group.

(1) If $|G| = \infty$, then $G \cong (\mathbb{Z}, +)$.

(2) If $|G| = n$, then $G \cong (\mathbb{Z}_n, +)$.

$$\begin{aligned}\phi : \mathbb{Z} &\rightarrow G \\ k &\mapsto a^k, \quad \forall k \in \mathbb{Z}\end{aligned}$$

$$\begin{aligned}\phi : \mathbb{Z}_n &\rightarrow G \\ k &\mapsto a^k, \quad \forall k \in \mathbb{Z}_n\end{aligned}$$

Theorem (Generators of Cyclic Groups)

Let $G = \langle a \rangle$ be a cyclic group.

- (1) If $|G| = \infty$, then a and a^{-1} are the only generators of G ;
- (2) If $|G| = n$, then a^r is a generator of G iff $(r, n) = 1$.

Suppose that $G = \langle a^k \rangle$.

$$\text{ord } a^r = \frac{n}{(n, r)}$$

$$\exists n \in \mathbb{Z} \ (a^k)^n = a^{kn} = a$$

$$kn = 1 \implies k = \pm 1$$

$$G = \langle a^r \rangle \iff \text{ord } a^r = n$$

$$\iff \frac{n}{(n, r)} = n$$

$$\iff (n, r) = 1$$

$$(\mathbb{Z}_{12}, +)$$

Generators : 1, 5, 7, 11

Theorem (Subgroups of Cyclic Groups)

Every subgroup of a cyclic group is cyclic.

Theorem (Subgroups of Cyclic Groups)

Suppose that $G = \langle a \rangle$ is a cyclic group.

(1) *If $|G| = \infty$, the set of subgroups of G is:*

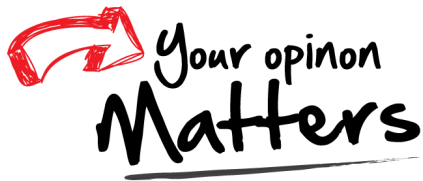
$$\{\langle a^d \rangle \mid d = 0, 1, 2, \dots\}$$

(2) *If $|G| = n$, the set of subgroups of G is:*

$$\{\langle a^d \rangle \mid d \text{ is a positive divisor of } n\}$$

$(\mathbb{Z}_{12}, +)$ has 6 subgroups

Thank
You!



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